

24CSA303 MOBILE APPLICATION DEVELOPMENT

Introduction to Android Studio

Android is a mobile operating system based on a modified version of the Linux kernel and other open source software, designed primarily for touchscreen mobile devices such as smartphones and tablets.



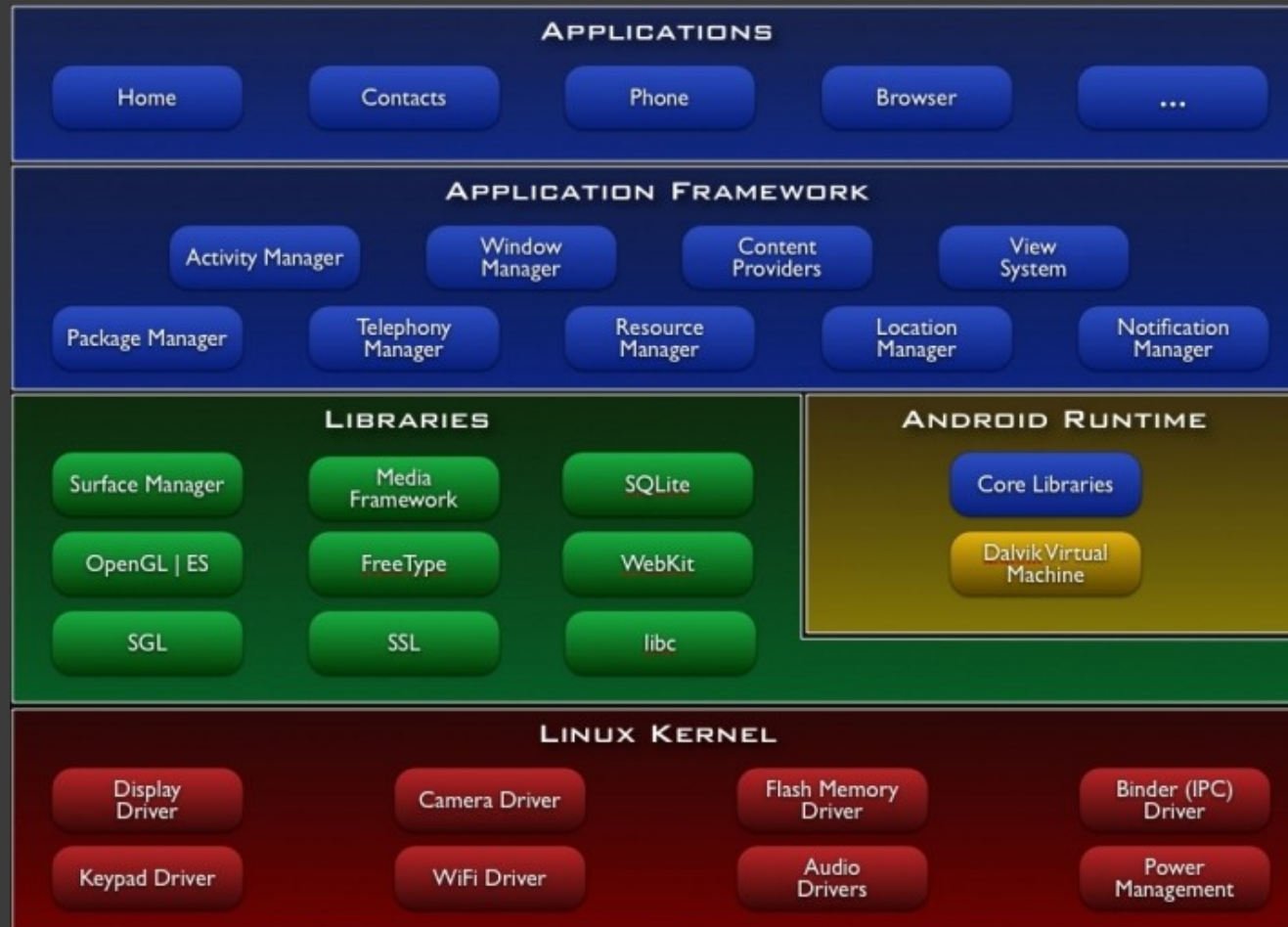
- **The Official IDE:** Android Studio is the standard, Google-backed development environment built on JetBrains' IntelliJ IDEA (Integrated Development Environment Application).
- software, *Which offers a powerful code editor and developer tool.*
- **Core Languages:** Projects are primarily written using Kotlin or Java for logic, while XML is utilized to design the user interface (UI).
- **System Minimums:** For stable presentation demonstrations, ensure devices have at least 4GB RAM (8GB+ recommended) and 2GB available disk space.



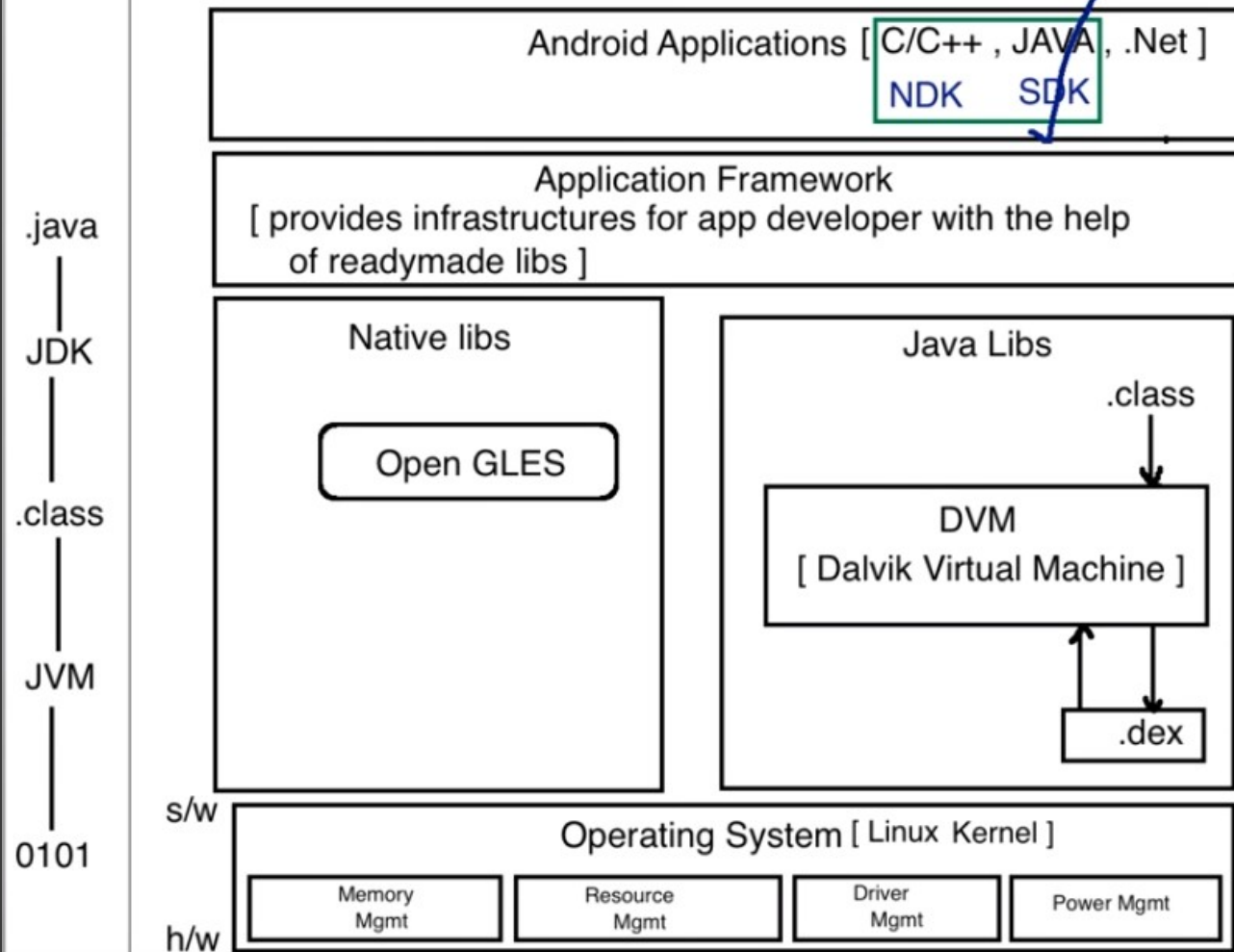
Andr.....oid

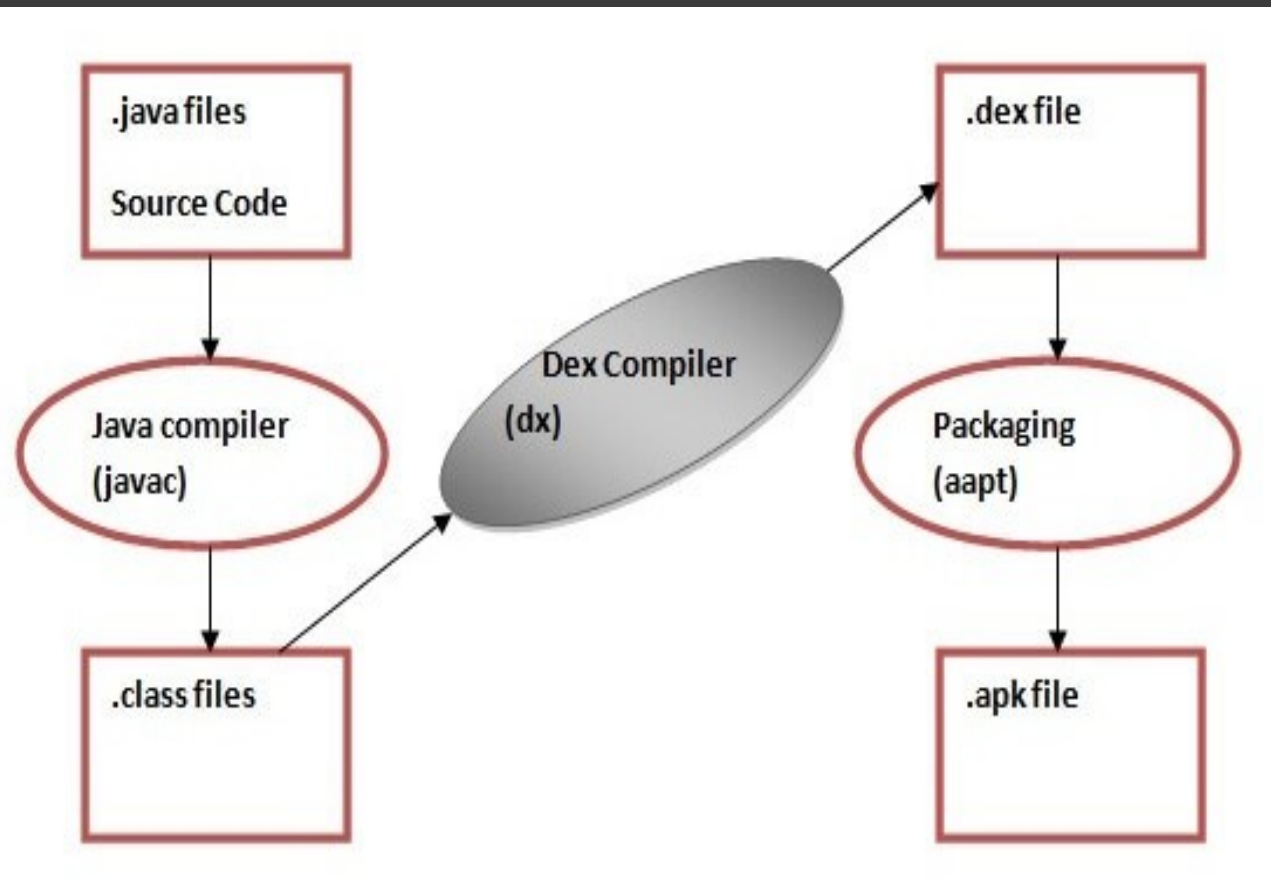
- The term **android** ultimately originates from the Greek words *andr-* (meaning "man" or "human") and *-oid* (meaning "resembling" or "like"). Together, the literal translation is "**man-like**" or "**human-like**"

Android System Architecture



Android Architecture





Android Versions

Names are according to a **DESERT** or **SUGARY FEAT**.

Version ↕	Code name ↕	Release date ↕	API level ↕	Distribution ↕
4.4	<i>KitKat</i>	October 31, 2013	19	8.5%
4.3.x	<i>Jelly Bean</i>	July 24, 2013	18	8.5%
4.2.x		November 13, 2012	17	18.8%
4.1.x		July 9, 2012	16	33.5%
4.0.3–4.0.4	<i>Ice Cream Sandwich</i>	December 16, 2011	15	13.4%
3.2	<i>Honeycomb</i>	July 15, 2011	13	0.1%
2.3.3–2.3.7	<i>Gingerbread</i>	February 9, 2011	10	16.2%
2.2	<i>Froyo</i>	May 20, 2010	8	1.0%

2.0 – 2.1 - Eclairs

1.6 - Doughnut

1.5 – Cupcake

1.1 – API 2

1.0 – API 1

API - Application
Programmable Interface

The first commercially available smartphone running Android was the **HTC DREAM**, released on **October 22, 2008** with Version **Cupcake(1.5)**



Recent Android Stable Version Timeline

Release	API Version	Released	Security Support
17 'Cinnamon Bun'	37	2 weeks and 1 day ago (16 Jun 2026)	Yes
16 'Baklava'	36	1 year ago (10 Jun 2025)	Yes
15 'Vanilla Ice Cream'	35	1 year and 10 months ago (03 Sep 2024)	Yes
14 'Upside Down Cake'	34	2 years and 9 months ago (04 Oct 2023)	Yes

Android Version	API Level	Release Date	Current Ecosystem Status
Android 17	API 37	June 16, 2026	Latest Release: Mainline version for new flagship hardware.
Android 16	API 36	June 10, 2025	Most Widely Used: Heavily active across consumer devices.
Android 15	API 35	September 3, 2024	Legacy Stable: Broadly deployed on mid-range/older devices.

Key Developer Features in the Latest Stable Release

- **Advanced Multitasking:** Implements system-wide floating "Bubbles" and desktop-style window orientation tools for tablets and foldable smartphones
- **Next-Gen Media Formats:** Built-in compatibility for **RAW14 image formats** and custom vendor camera extension tools.
- **Eclipsa Video Standard:** Natively processes HDR video streams while adapting dynamic lighting directly to real-world ambient environment sensors.
- **Granular Privacy Controls::** Features single-use temporary location boundaries and specialized security modules that lock hardware immediately if a theft pattern is detected

What This Means For Your Android Studio Project

- Set your `compileSdk` to **37** to compile with the latest stable APIs.
- Set your `targetSdk` to **37** to tell the operating system that your app is thoroughly tested for Android 17 behaviors.
- Set your `minSdk` to **35 or 36** if you want to ensure your application can still run on the largest portion of devices globally without breaking legacy functionality.

The 4 Structural Pillars of Android Apps

- **Activities:** The visual entry points representing a single user-interface screen.
- **Services:** Background processes that run without providing a direct user interface.
- **Broadcast Receivers:** Gateway elements that allow your app to respond to system-wide announcement signals.
- **Content Providers:** Managed data structures used to safely share data between different apps

Essential Project File Architecture

- **app/manifests/**[AndroidManifest.xml](#): Every app must have this file at its root to declare permissions, activities, and fundamental components to the Android system.
- **app/java/**: Houses the Kotlin or Java source code files (e.g., MainActivity.kt).
- **app/res/layout/**: Contains the visual structural design files written in XML format.
- **Gradle Scripts**: Automation build files that manage external plugins, project dependencies, and software versioning.

test1 ▾

Version control ▾

Pixel 8 ▾

app ▾

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Android ▾

📁 app

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📄 Gradle Scripts

📄 build.gradle.kts (Project: test1)

📄 build.gradle.kts (Module :app)

📄 proguard-rules.pro (ProGuard R

📄 gradle.properties (Project Prop

📄 gradle-wrapper.properties (Grad

📄 libs.versions.toml (Version Cata

📄 local.properties (SDK Location)

📄 settings.gradle.kts (Project Sett

Search Everywhere Double Shift

Go to File Ctrl+Shift+N

Recent Files Ctrl+E

Navigation Bar Alt+Home

Drop files here to open them

Running Device

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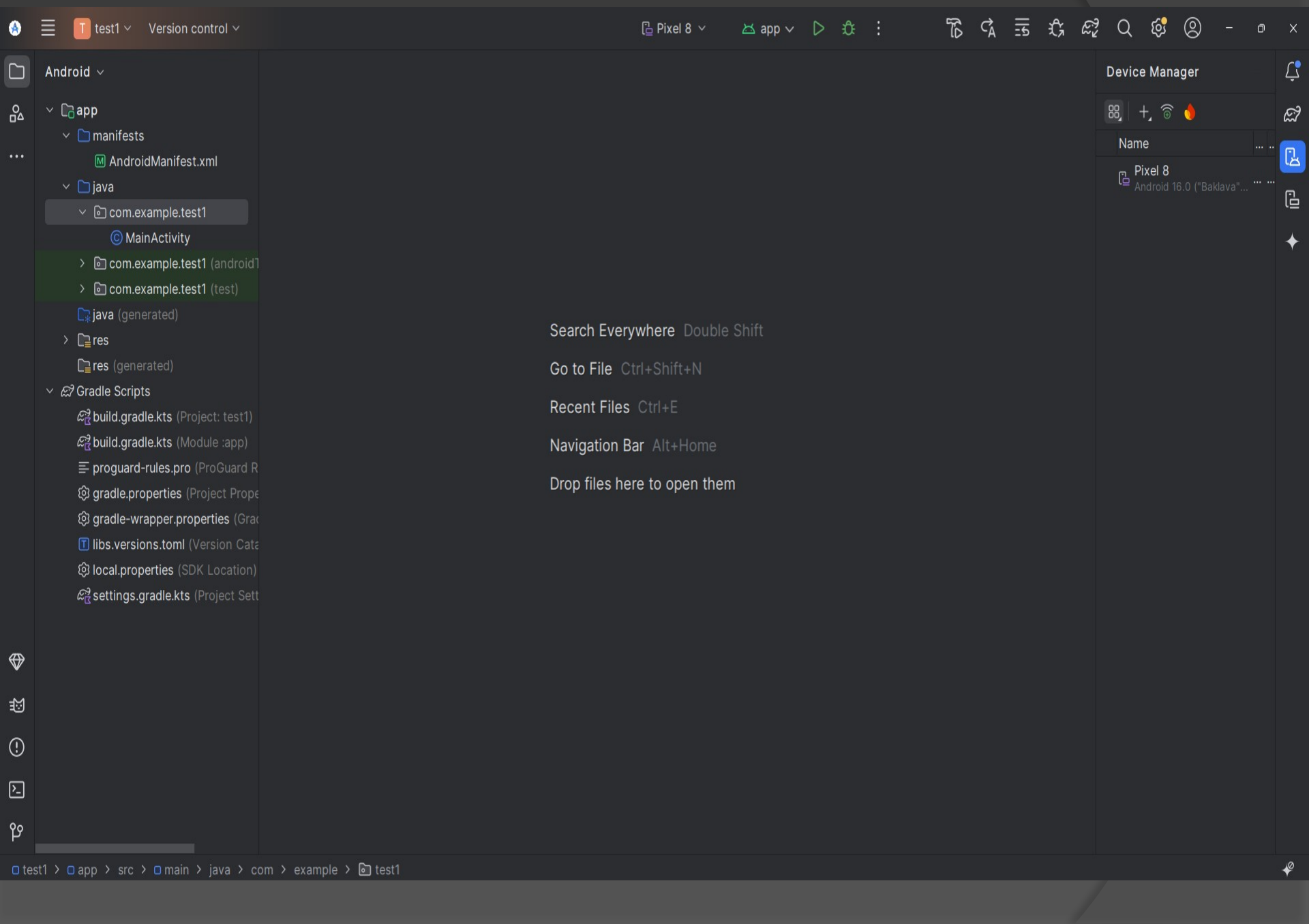
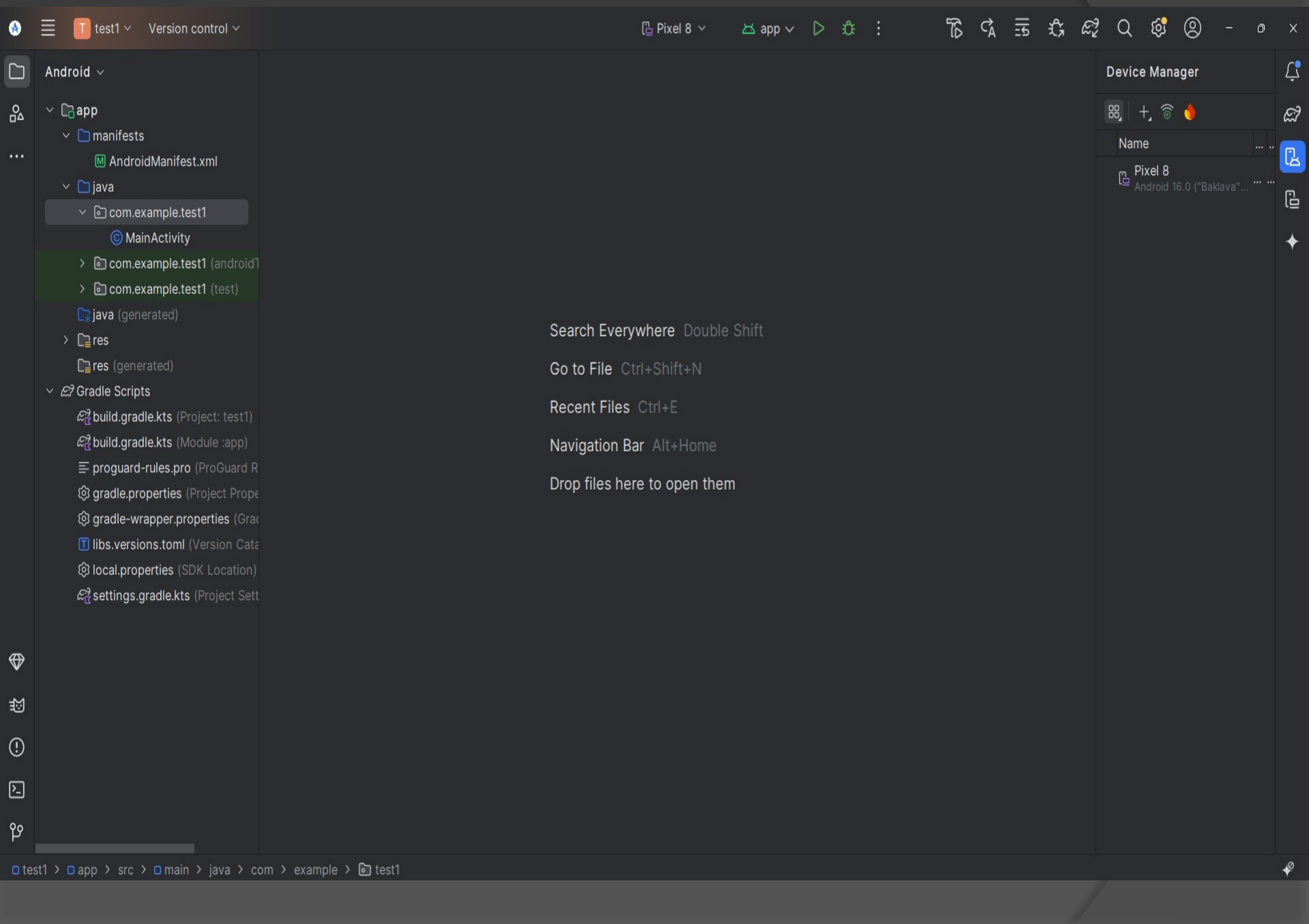
★



To mirror a physical device, connect it via USB cable or over WiFi, click **+** and select the device from the list. You may also select the **Activate mirroring when a new physical device is connected** option in the [Device Mirroring settings](#).

To launch a virtual device, click **+** and select the device from the list, or use the [Device Manager](#).

test1 > .gradle > config.properties



test1 ▾ Version control ▾

Android ▾

app

manifests

AndroidManifest.xml

java

com.example.test1

MainActivity

com.example.test1 (android)

com.example.test1 (test)

java (generated)

res

drawable

ic_launcher_background.xml

ic_launcher_foreground.xml

layout

menu

mipmap

values

colors.xml

new_launcher_background.xml

strings.xml

style1.xml

themes (2)

xml

res (generated)

Gradle Scripts

build.gradle.kts (Project: test1)

build.gradle.kts (Module :app)

proguard-rules.pro (ProGuard R

gradle.properties (Project Prop

gradle-wrapper.properties (Grad

Search Everywhere Double Shift

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Pixel 8 API 36

Pixel 8 API 36

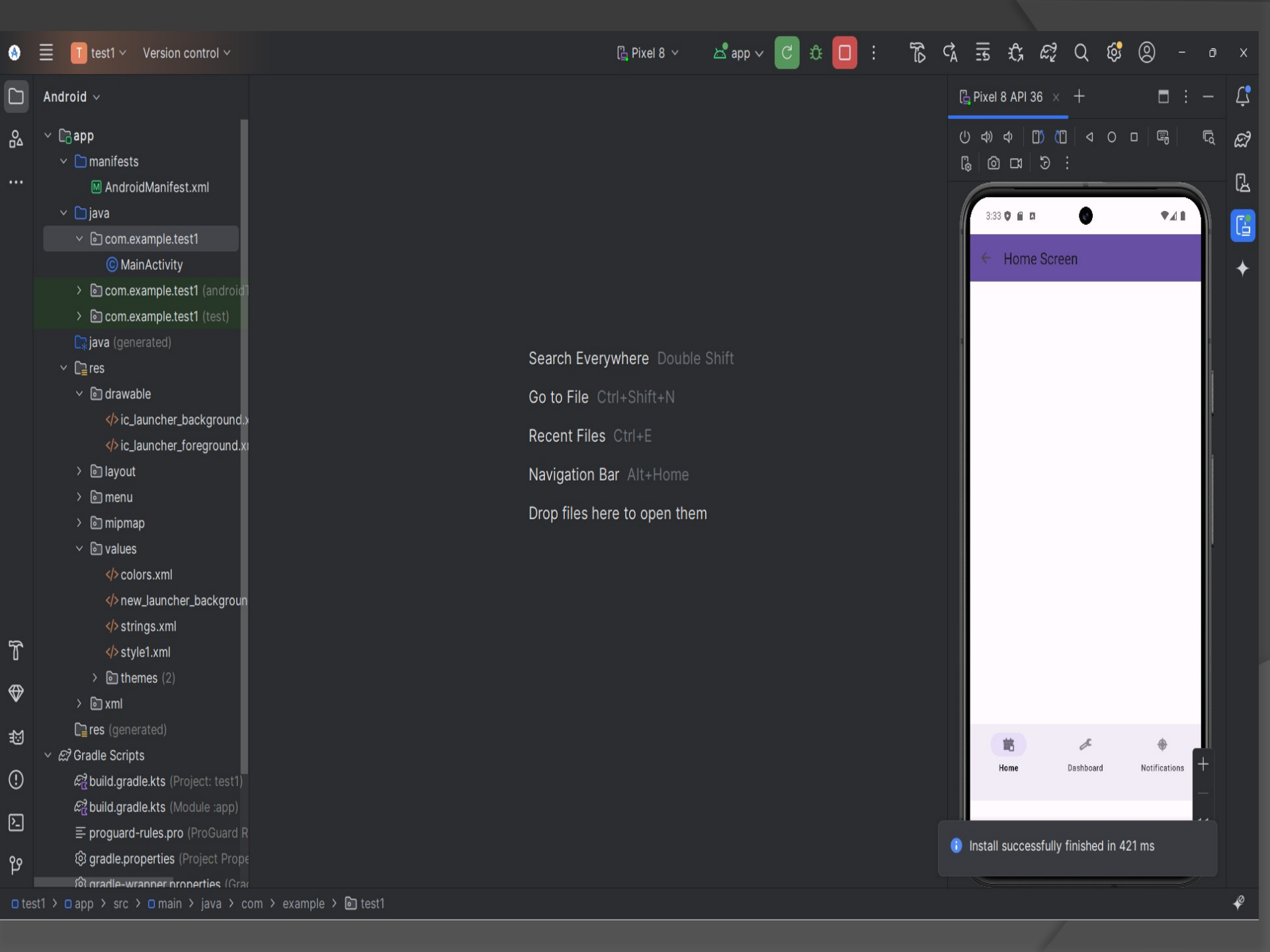
3:33

Home Screen

Home Dashboard Notifications

Install successfully finished in 421 ms

test1 > app > src > main > java > com > example > test1



Questions.....

1. Which directory in an Android Studio project contains the resource files like XML layouts, images, and string values? 

A. java/

B. res/

C. manifests/

D. gradle/

Hide hint 

Think about where you would look to find the visual layout file (e.g., `activity_main.xml`) or the app's logo icon.

2. Which XML attribute is universally used to assign a unique identifier to a UI component so it can be referenced in your Java code? 

A. `android:id`

B. `android:layout_width`

C. `android:name`

D. `android:tag`

Hide hint 

When you define an element in XML, you write it in the format of `@+id/your_custom_name` .

3. Which callback method in the Android Activity lifecycle is called first when an Activity is starting, and is where you typically call `setContentView()` ?

A. `onStart()`

B. `onResume()`

C. `onCreate()`

D. `onPause()`

Hide hint ^

Think of this method as the "birth" or "constructor" of the Activity itself, where you inflate the user interface.

4. In Java, which method is explicitly used to retrieve a reference to a UI component (like a TextView or Button) from the XML layout?

A. `getView()`

B. `findViewById()`

C. `findViewWithId()`

D. `inflateLayout()`

Hide hint ^

If you want to access a Button whose ID is `R.id.my_submit_button` , you would pass this ID as a parameter to this specific method.

5. Which basic UI component is primarily designed to allow the user to input and edit text in an Android application?

A. TextView

B. Button

C. ImageView

D. EditText

Hide hint ^

If a user needs to type their username or password, which of these components is the best fit?

6. Immediately after an Activity's `onCreate()` method finishes, what state does the Activity enter next, causing it to become visible to the user? 

A. `onStart()`

B. `onResume()`

C. `onStop()`

D. `onDestroy()`

Hide hint 

The lifecycle callback that makes the activity visible occurs just before `onResume()` .

7. To make a Button respond to a user's tap in Java, which method is typically called on the Button object to set up an event listener?

A. `setOnTouchListener()`

B. `setOnClickListener(null)`

C. `setOnClickListener()`

D. `setClickAction()`

Hide hint ^

What are you listening for when the user "clicks" or "taps" a button?

8. Which layout type in Android organizes its child views in a single direction, either strictly horizontally or vertically? 

A. `ConstraintLayout`

B. `LinearLayout`

C. `RelativeLayout`

D. `FrameLayout`

Hide hint 

Think of the word that describes placing items in a single, straight row or column.

9. Which XML attribute is used to define the actual text displayed inside a `TextView` or `Button`? 

A. `android:text`

B. `android:hint`

C. `android:contentDescription`

D. `android:title`

Hide hint 

If you want a button to display "Submit" or a `TextView` to display "Hello World", you use an attribute starting with `android:t...`

10. In an Android Studio project, which file is primarily responsible for configuring app-level build settings, module dependencies, and the SDK versions?

A. AndroidManifest.xml

B. strings.xml

C. MainActivity.java

D. build.gradle

Hide hint ^

Android Studio uses a specific build system that requires you to define your external libraries (dependencies) inside files with the `.gradle` extension.

11.

WAP stands for?



Wireless Application Protocol



Wireless Application Program



Wireless Application Portal



Wireless Application Parts

12

In android, mini activities are also known as

- ☐ Adapter
- ☐ Activity
- ☒ Fragments
- ☐ None

13

In which of the following is the Android application Framework provided?

☐ Zip file

☒ Jar file

☐ Object file

☐ EXE file

JSON stands for _____

- ☐ JavaScript Oriented Notation
- ☐ JavaScript Object Native
- ☒ JavaScript Object Notation
- ☐ None

15

OHA stands for _____.



Open Handset Alliance



Open Handset Association



Open Handset Application



Open Host Application

16

On what is Android web browser-based?

- ☐ Safari
- ☐ Firefox
- ☒ Open-source WebKit
- ☐ Chrome

17

On which kernel is Android-based on

☐ Windows

☐ Mac

☒ Linux

☐ Redhat

18

The full form of ADB is.

- ☐ Android Delete Bridge
- ☒ **Android Debug Bridge**
- ☐ Android Destroy Bridge
- ☐ Android Developed Bridge

19

The full form of ANR is.

- ☐ Application not rendering
- ☒ Application not responding
- ☐ Application not reacting
- ☐ Application not reachable

The full form of GCM is

- ☐ Google Centre messaging
- ☒ Google Cloud messaging
- ☐ Google account
- ☐ None account manager

Identify the topmost layer of Android architecture

- ☐ Applications
- ☒ Applications Framework
- ☐ Linux Kernel
- ☐ System libraries and android runtime

. What is an activity in android?

☐

android class

☐

android package

☒

A single screen in an application with supporting java code

☐

None of the above

23

What is manifest XML in android?

- ☐ It has information about layout in an application
- ☒ It has all the information about an application
- ☐ It has the information about activities in an application
- ☐ None

24

Who develops DVM?



Dan Bornstein



Dennis Ritchie



Linus Torvald



None

5. Who manages data sharing between applications?

☐ Telephony managers

☐ Location manager

☒ Content providers

☐ Activity manager

- <https://www.youtube.com/watch?v=JAsdcM0PPn4>
- <https://developer.android.com/courses/android-basics-compose/course>

Thank You